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Rotor for an electric machine

Abstract

An interior permanent magnet (IPM) electric machine has an improved rotor configuration to manage mechanical stresses induced by electro-magnetic force acting upon permanent magnets housed therein. This includes providing magnet cavities in the rotor with sufficient clearances in the corners wherein a portion of a slot corner is formed with certain curvature shapes using a novel geometry. By doing this, more surface area is obtained to evenly distribute stress that is induced by centrifugal force acting upon the rotor during rotation, thus reducing the stress concentration. Furthermore, an expanded space is achieved between the magnet corner and the rotor lamination, thus providing robust packaging and dynamic support of the permanent magnets in the magnet cavities. Furthermore, the expanded space provides improved clearance for ease of manufacturing and assembly.

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Background/Summary

INTRODUCTION

(1) Rotary electric machines, such as motor-generator units, include a rotor configured to rotate about a shaft defining an axis, and a stator. One type of the rotary electric machine is an interior permanent magnet (IPM) electric machine, wherein the rotor includes a plurality of permanent magnets that are arranged in longitudinal slots that are formed near its outer periphery. The plurality of permanent magnets interact with rotating magnetic fields generated by the stator to induce torque on the shaft of the rotor to urge its rotation.

(2) During rotation, a rotor experiences mechanical stress as electro-magnetic force generated via a stator acts upon the plurality of permanent magnets to generate torque that is transferred to the rotor

shaft, with the mechanical stress being caused by centrifugal force of the permanent magnets in the longitudinal slots of the rotor. The dynamic speed and torque operating ranges of the electric machine may be limited based upon the mechanical stresses.

(3) However, engineers seek to design electric motors that are capable of higher speed operation to increase power density and reduce the size of the motor. As motor speed increased, it has been found that known fillet designs in magnet slot corners that worked well at medium to low speeds actually limit ability to achieve very high-speed operation due to stress concentration that is induced by centrifugal force.

(4) Known designs for reducing stress concentration at the magnet slot corners include use of fillets by adding (or not removing) inside corner material towards the magnet, with larger corner radii being preferred to achieve improved stress reduction. However, for motors at very high speeds, target stress values are more difficult to achieve due to space limitations, and may result in a smaller clearance between the magnet and the rotor. This may lead to issues during manufacturing and assembly, including part-to-part variability, tolerances, etc.

(5) There is a need for an improved rotor configuration for an IPM electric machine to better manage mechanical stresses caused by the centrifugal force of the permanent magnets acting upon the rotor. There is also need for an improved rotor configuration to provide improved robustness of a rotor for an IPM electric machine, including robustness to placement of permanent magnets within longitudinal slots that are formed in the rotor. There is also need for improved manufacturability of a rotor, including an improved ability to insert permanent magnets within longitudinal cavities that are formed in the rotor.

SUMMARY

(6) The concepts described herein provide an interior permanent magnet (IPM) electric machine having an improved rotor configuration to better manage mechanical stresses induced by the electro-magnetic force acting upon the permanent magnets housed therein. This includes providing magnet cavities in the rotor with sufficient clearances in the corners wherein a portion of a slot corner is formed with certain curvature shapes using a novel geometry. By doing this, more surface area is obtained to evenly distribute stress that is induced by centrifugal force at high-speed rotation, thus reducing the stress concentration. Furthermore, an expanded space is achieved between the magnet corner and the rotor lamination, thus providing robust packaging and dynamic support of the permanent magnets in the magnet cavities. Furthermore, the expanded space provides improved clearance for ease of manufacturing and assembly.

(7) An aspect of the disclosure includes a laminate plate for a rotor of an interior permanent magnet electric machine. The laminate plate is a disk-shaped plate having a plurality of elongated apertures, wherein each of the elongated apertures has opposed first and second sides, and opposed first and second ends. The first end is disposed towards an axis of rotation of the disk-shaped plate, and the second end is disposed towards an outer periphery of the disk-shaped plate. The second end includes a corner portion and a load bearing portion, wherein the load bearing portion is arranged perpendicular to the first side. The corner portion has an edge that defines a void area that is recessed into the first side. The edge has a non-linear geometry that includes a convex portion arranged in series with a concave portion, wherein the convex portion is connected to the first side and the concave portion is connected to the load bearing portion of the second end. The edge minimizes a stress concentration in the corner portion that may be induced by centrifugal force of an element such as a permanent magnet that is disposed in the elongated aperture and is acting upon the load bearing portion when the rotor is rotating in the IPM electric machine.

(8) Another aspect of the disclosure includes the convex portion of the edge of the corner portion having a first axis of rotation **R1** and the concave portion having a second axis of rotation **R2**, wherein the first axis of rotation **R1** is greater than the second axis of rotation **R2**.

(9) Another aspect of the disclosure includes the convex portion of the edge of the corner portion having a first axis of rotation **R1** and the concave portion having a second axis of rotation **R2**,

wherein the first axis of rotation R1 is less than 2 mm, and wherein the second axis of rotation R2 is greater than 0.05 mm.

(10) Another aspect of the disclosure includes the edge of the corner portion being a complex spline curve having maximum depth into the plate that is less than 2 mm.

(11) Another aspect of the disclosure includes the edge minimizing the stress concentration in the corner portion that is induced by centrifugal force of the element that is disposed in the aperture when the rotor is rotating in the IPM electric machine, wherein the element is arranged to act upon the load bearing portion.

(12) Another aspect of the disclosure includes the laminate plate being fabricated from electrical steel.

(13) Another aspect of the disclosure includes the plurality of apertures being circumferentially arranged.

(14) Another aspect of the disclosure includes the plurality of laminations being disk-shaped laminations that are disposed on the rotatable shaft in a longitudinally-stacked arrangement, wherein each of the laminations includes a plurality of apertures, wherein the laminations are arranged on the rotatable shaft such that the apertures are aligned to form the axially-disposed cavities.

(15) Another aspect of the disclosure includes a rotor for an interior permanent magnet (IPM) electric machine that includes a rotatable shaft disposed on a longitudinal axis, a plurality of disk-shaped plates arranged on the rotatable shaft, and a plurality of permanent magnets. Each of the plurality of disk-shaped plates defines a plurality of apertures. The plurality of disk-shaped plates are arranged with the plurality of apertures being aligned to form a plurality of axially-disposed cavities, and the plurality of permanent magnets are disposed in the plurality of axially-disposed cavities. Each of the apertures of the plurality of disk-shaped plates has opposed first and second sides, and opposed first and second ends, wherein the first end is disposed towards the longitudinal axis, and wherein the second end is disposed towards an outer periphery of the disk-shaped plate. The second end includes a corner portion and a load bearing portion, and the load bearing portion is arranged perpendicular to the first side. The corner portion has an edge that defines a void area that is recessed into the first side, wherein the edge includes a convex portion arranged in series with a concave portion. The convex portion is connected to the first side, and the concave portion is connected to the load bearing portion of the second end. The edge minimizes a stress concentration in the corner portion that is induced by centrifugal force of a respective one of the permanent magnets that is disposed in the aperture when the rotor is rotating.

(16) Another aspect of the disclosure includes an interior permanent magnet (IPM) electric machine that includes a rotor that is rotatably arranged within a stator. The rotor includes a rotatable shaft, disposed on a longitudinal axis, a plurality of laminate plates arranged on the rotatable shaft, and a plurality of permanent magnets. Each of the plurality of laminate plates defines a plurality of apertures. The plurality of disk-shaped plates are arranged with the plurality of apertures being aligned to form a plurality of axially-disposed cavities, and the plurality of permanent magnets are disposed in the plurality of axially-disposed cavities. Each of the apertures of the plurality of disk-shaped plates has opposed first and second sides, and opposed first and second ends, wherein the first end is disposed towards the longitudinal axis, and wherein the second end is disposed towards an outer periphery of the disk-shaped plate. The second end includes a corner portion and a load bearing portion, and the load bearing portion is arranged perpendicular to the first side. The corner portion has an edge that defines a void area that is recessed into the first side, wherein the edge includes a convex portion arranged in series with a concave portion. The convex portion is connected to the first side, and the concave portion is connected to the load bearing portion of the second end. The edge minimizes a stress concentration in the corner portion that is induced by centrifugal force of a respective one of the permanent magnets that is disposed in the aperture when the rotor is rotating in the stator.

(17) The above features and advantages, and other features and advantages, of the present teachings are readily apparent from the following detailed description of some of the best modes and other embodiments for carrying out the present teachings, as defined in the appended claims, when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) One or more embodiments will now be described, by way of example, with reference to the accompanying drawings, in which:
- (2) FIG. 1 schematically illustrates a system that includes a multi-phase rotary interior permanent magnet (IPM) electric machine (“electric machine”) that is arranged to generate and transfer torque to an actuator to effect work, in accordance with the disclosure.
- (3) FIG. 2 schematically shows a cutaway side-view of a stack of laminate plates for a rotor of an IPM electric machine, in accordance with the disclosure;
- (4) FIG. 3 schematically shows an end-view a laminate plate for a rotor of an IPM electric machine, in accordance with the disclosure.
- (5) FIGS. 3A, 3B, 3C, and 3D schematically show details related to embodiments of the laminate plate for a rotor of an IPM electric machine that is described with reference to FIG. 3, in accordance with the disclosure.
- (6) FIG. 4 schematically illustrates an end-view of a portion of a known embodiment of a laminate plate for a rotor of an IPM electric machine, including a magnitude of induced stress superimposed thereon, in accordance with the disclosure.
- (7) FIG. 5 schematically illustrates an end-view of a portion of an embodiment of a laminate plate for a rotor of an embodiment of the IPM electric machine described herein, including a magnitude of induced stress superimposed thereon, in accordance with the disclosure.
- (8) FIG. 6 graphically illustrates maximum rotor stress for a known system and an embodiment of the rotor of the IPM electric machine described herein, in accordance with the disclosure.
- (9) The appended drawings are not necessarily to scale, and may present a somewhat simplified representation of various preferred features of the present disclosure as disclosed herein, including, for example, specific dimensions, orientations, locations, and shapes. Details associated with such features will be determined in part by the particular intended application and use environment.

DETAILED DESCRIPTION

(10) The components of the disclosed embodiments, as described and illustrated herein, may be arranged and designed in a variety of different configurations. Thus, the following detailed description is not intended to limit the scope of the disclosure, as claimed, but is merely representative of possible embodiments thereof. In addition, while numerous specific details are set forth in the following description in order to provide a thorough understanding of the embodiments disclosed herein, some embodiments can be practiced without some of these details. Moreover, for the purpose of clarity, certain technical material that is understood in the related art has not been described in detail in order to avoid unnecessarily obscuring the disclosure. For purposes of convenience and clarity only, directional terms such as top, bottom, left, right, up, over, above, below, beneath, rear, and front, may be used with respect to the drawings. These and similar directional terms are not to be construed to limit the scope of the disclosure. Furthermore, the disclosure, as illustrated and described herein, may be practiced in the absence of an element that is not specifically disclosed herein.

(11) Referring to the drawings, wherein like reference numbers refer to the same or like components in the several Figures, FIG. 1 schematically illustrates a system **100** that includes an electric machine **110** that is arranged to generate and transfer torque to an actuator **140** to effect

work, and a controller **130** to control and manage operation thereof. In one embodiment, the system **100** is disposed on a vehicle (not shown). When disposed on a vehicle, the vehicle may include, but not be limited to a mobile platform in the form of a commercial vehicle, industrial vehicle, agricultural vehicle, passenger vehicle, aircraft, watercraft, train, all-terrain vehicle, personal movement apparatus, robot and the like to accomplish the purposes of this disclosure. The electric machine **110** may be configured as an electric motor that is capable of transforming electric power to mechanical torque, a generator that is capable of transforming mechanical torque to electric power, or as a motor/generator that is capable of both.

(12) The system **100** also includes an inverter **104** and a DC power source **102**. The DC power source **102** connects to the inverter **104** via a high-voltage DC bus **103**, and the inverter **104** connects to the electric machine **110** via a plurality of electrical power lines **106**.

(13) The electric machine **110** is configured as a multi-phase rotary interior permanent magnet (IPM) electric machine (“electric machine”) **110** having a stator **112** and a rotor **114**, wherein the rotor **114** is arranged on a rotor shaft **116** and is coaxial with and rotatable within the stator **112**.

(14) The inverter **104** includes a plurality of semiconductor switches (not shown) that are arranged and controllable to transform DC electric power to AC electric power, and transform AC electric power to DC electric power, employing a pulse-width modulation signal **108** or another control technique. The inverter **104** is arranged and controllable to transform DC electric power originating from the DC power source **102** to AC electric power to actuate the stator **112** of the electric machine **110** to urge the rotor **114** to rotate and generate mechanical torque that is transferred via a rotatable member **120** and a geartrain **122** to the actuator **140** when operating in a torque generating mode. The electric machine **110** is controllable to generate AC electric power from mechanical torque originating at the actuator **140**, which is transformed by the inverter **104** to DC electric power for storage in the DC power source **102** when operating in an electric power generating mode. The actuator **140** includes, in one embodiment, a vehicle wheel that transfers torque to a ground surface to effect forward motion as part of a traction propulsion system. The DC power source **102** may be in the form of a rechargeable electrochemical battery device, a fuel cell, an ultracapacitor, and/or another battery technology.

(15) Sensors are arranged to monitor parameters of the system **100**. Monitored parameters include, by way of non-limiting examples, voltage and current between the electric machine **110** and the inverter **104**, and rotational speed of the electric machine **110**. Other monitored parameters may include, e.g., voltage at the high-voltage DC bus **103**, etc. Other monitored parameters may include motor commands, including a motor torque command, which may originate from an operator torque request or a system request associated with the actuator **140**.

(16) The controller **130** may be embodied as one or more digital computing devices, and may include one or more processors **134** and sufficient amounts of tangible non-transitory memory **132**. A control routine may be stored as an executable instruction set in the memory **132** and executed by one of the processors **134** of the controller **130**. The controller **130** is in communication with the inverter **104** and the aforementioned sensors to control operation thereof in response to execution of the control routine to operate the electric machine **110**.

(17) The term “controller” and related terms such as microcontroller, control module, module, control, control unit, processor and similar terms refer to one or various combinations of Application Specific Integrated Circuit(s) (ASIC), Field-Programmable Gate Array (FPGA), electronic circuit(s), central processing unit(s), e.g., microprocessor(s) and associated non-transitory memory component(s) in the form of memory and storage devices (read only, programmable read only, random access, hard drive, etc.).

(18) FIGS. 2 and 3 schematically illustrate a side-view and an end-view, respectively, of a laminate stack **200** for an embodiment of the rotor **114** of the IPM electric machine **110** that is described with reference to FIG. 1.

(19) FIG. 2 schematically shows a cutaway side-view of the laminate stack **200**, which is composed

of a plurality of laminate plates **210** that may be arranged as elements that are mounted on an embodiment of the rotor shaft **116** of the rotor **114** that is described with reference to FIG. 1. FIG. 2B schematically shows a corresponding end-view one of the laminate plates **210**. A rotor longitudinal axis **201** and a radial axis **202** are defined by the laminate plate **210**. The rotor longitudinal axis **201** defines an axis of rotation of the rotor **114** and rotor shaft **116**.

(20) Each of the laminate plates **210** is arranged as a circular disk that has an inner or center aperture **214** and outer periphery **216**. Each of the laminate plates **210** has a plurality of elongated apertures **230** that are circumferentially arranged and formed therein. When a plurality of the laminate plates **210** are arranged in a stack, as shown with reference to FIG. 2, the plurality of elongated apertures **230** are radially aligned to form a plurality of cavities **220** that are parallel with the rotor longitudinal axis **201**. Permanent magnets **218** are inserted into and secured in the plurality of axially-disposed cavities **220**. Each of the elongated apertures **230** defines a longitudinal aperture axis **235**.

(21) As shown, the plurality of elongated apertures **230** are arranged in aperture sets. Aperture longitudinal axes **235** of the elongated apertures **230** are placed at angles **236** in relation to radial axes **202** that are symmetrical about a respective one of the axes **202**. As shown, there are eight sets of the elongated apertures **230**.

(22) Each of the laminate plates **210** is formed from a ferromagnetic material, e.g., electrical steel, which is an iron alloy tailored to produce specific magnetic properties that may include a small hysteresis area resulting in low power loss per cycle, low core loss, and high permeability. Electrical steel is an iron alloy which may have from zero to 6.5% silicon (Si:5Fe). Commercial alloys may have silicon content up to 3.2%. Manganese and aluminum can be up to 0.5%. In one embodiment, the electrical steel is a grain-oriented electrical steel.

(23) Referring now to FIG. 3 with continued reference to the IPM electric machine **110** of FIG. 1, a plate section **211** of an embodiment of the laminate plate **210** for the rotor **114** is described. The plate section **211** is arranged between center aperture **214** and outer periphery **216**, and includes a subset of a set **225** of the elongated apertures **230**. The plate section **211** defines a first elongated aperture **230** that is arranged in parallel with a second elongated aperture **229**, wherein both define a longitudinal aperture axis **235**, and both are placed such that their respective longitudinal aperture axis **235** is at an acute angle **236** in relation to radial axis **202**.

(24) In one embodiment, each of the first elongated apertures **230** is configured to accommodate a portion of one of the permanent magnets **218**, which may be arranged as a rectangular prism having a rectangular cross-section.

(25) The first elongated aperture **230** has a first side **231** that is opposed to a second side **232**, with a first end **233** opposed to a second end **234**. The first end **233** is disposed towards the axis of rotation of the laminate plate **210**, i.e., rotor longitudinal axis **201**, and the second end **234** is disposed towards the outer periphery **216**. The second end **234** includes a corner portion **240** and a load bearing portion **245**. The load bearing portion **245** is arranged perpendicular to the first side **231** (and to the second side **232** in one embodiment and as shown).

(26) Referring now to FIGS. 3B and 3C, with continued reference to the IPM electric machine **110** of FIG. 1, an expanded view of a portion of the laminate plate **210** is shown, including the corner portion **240** and the load bearing portion **245** that define a portion of the first elongated aperture **230**. The corner portion **240** has an edge **241** that defines a void area **244** that is recessed into the first side **231** of the laminate plate **210**. The edge **241** has a convex portion **242** that is arranged in series with a concave portion **243**, with the convex portion **242** being connected to the first side **231** and the concave portion **243** being connected to the load bearing portion **245** of the second end **234**. As illustrated with reference to FIG. 3C, the edge of the corner portion **240** includes the convex portion **242** having a first axis of rotation with a first radius R1 and the concave portion **243** having a second axis of rotation with a second radius R2, wherein the first radius R1 is less than 2 mm, and wherein the second radius R2 is greater than 0.05 mm. Furthermore, a first line L1 is

defined by the first side **231**, and a second line **L2** is defined as being parallel to the first line **L1** and at a depth **d1** that is recessed into the first side **231** of the laminate plate **210**. Depth **d1** is greater than 0.0, but is less than 2 mm. The first radius **R1**, the second radius **R2**, and the depth **d1** are designed to minimize a stress concentration in the corner portion **240** that is induced by centrifugal force of an element that is disposed in the elongated aperture **230** during rotation. Example data indicating stress concentrations in an embodiment of a corner portion of an aperture of a laminate plate is illustrated with reference to FIG. 5. In one embodiment, the element is a portion of permanent magnet **218**, which is disposed in cavity **220** that is formed by alignment of the plurality of elongated apertures **230** in the plurality of the laminate plates **210** that are assembled into laminate stack **200** as part of rotor **114** that rotates in IPM electric machine **110**.

(27) Referring now to FIGS. 3B and 3D, with continued reference to the IPM electric machine **110** of FIG. 1, an expanded view of a portion of another embodiment of the laminate plate **210'** is shown, including the corner portion **240'** and the load bearing portion **245** that define a portion of the first elongated aperture **230'**. The corner portion **240'** has an edge **341** that defines a void area **344** that is recessed into the first side **231** of the laminate plate **210**. The edge **341** has a convex portion **342** that is arranged in series with a concave portion **343**, with the convex portion **342** being connected to the first side **231** and the concave portion **343** being connected to the load bearing portion **245** of the second end **234**. As illustrated with reference to FIG. 3D, the edge **341** of the corner portion **240** includes the convex portion **342** and the concave portion **343** being a complex spline curve having maximum depth into the plate that is less than 2 mm, wherein the maximum depth is measured by first line **L1** defined by the first side **231**, and second line **L2** that is parallel to the first line **L1** and at a depth **d1** that is recessed into the first side **231** of the laminate plate **210**. Depth **d1** is greater than 0.0, but is less than 2 mm. The complex spline curve of edge **341** is designed to minimize a stress concentration in the corner portion **240** that is induced by centrifugal force of an element that is disposed in the elongated aperture **230'** during rotation. Example data indicating stress concentration in an embodiment of a corner portion of an aperture of a laminate plate is illustrated with reference to FIG. 5. In one embodiment, the element is a portion of permanent magnet **218**, which is disposed in a cavity **220** that is formed by alignment of a plurality of elongated apertures **230** in a plurality of the laminate plates **210** that are assembled into rotor stack **200** as part of rotor **114**. As appreciated, a spline curve is a piecewise polynomial curve that is generated employing curve fitting and interactive curve design. As employed in design of the edge **341** of the corner portion **240**, the complex spline curve of edge **341** may be generated by inducing a force, e.g., centrifugal force, on the load bearing portion **245** and iteratively selecting and testing, in a piecewise manner, curve segments that minimize stress in the corner portion **240** employing computer-aided engineering methods.

(28) FIG. 4 schematically illustrates a computer-aided engineering (CAE) diagram **400** of a portion of an embodiment of a known laminate plate **410** of a rotor for an IPM electric machine, including a second end **434** of an elongated aperture **430** with a known 90 degree fillet corner portion **440** and load bearing portion **445**. A permanent magnet **418** is also shown. The CAE diagram **400** shows magnitudes of stress on the laminate **410** that is induced by rotation of the rotor in a stator of the IPM electric machine. Relative magnitudes of stress **450** are depicted, ranging between a maximum stress level **452** and a minimum stress level **454**. A first stress concentration **456** occurs at a point of contact between the load bearing portion **445** and the permanent magnet **418**. Furthermore, a second stress concentration **458** occurs in the fillet corner portion **440**. The first and second stress concentrations **456**, **458** are at the maximum stress level **452**.

(29) FIG. 5 schematically illustrates a computer-aided engineering (CAE) diagram **500** of a portion of an embodiment of a laminate plate **510** of a rotor for an IPM electric machine that has been designed in accordance with the concepts described with reference to FIGS. 3A-3D. This includes a second end **534** of an elongated aperture **530** with a corner portion **540** and load bearing portion **545**, wherein the corner portion **540** includes an embodiment of edge **541** with a convex portion

542 that is arranged in series with a concave portion 543, with the convex portion 542 being connected to the first side 532 and the concave portion 543 being connected to the load bearing portion 545 of the second end 534. The edge 541 is arranged as a complex spline curve. A permanent magnet 518 is also shown. The CAE diagram 500 shows magnitudes of stress on the laminate plate 510 that is induced by rotation of the rotor in a stator of the IPM electric machine. Relative magnitudes of stress 550 are depicted, ranging between a maximum stress 552 and a minimum stress 554. A first stress concentration 556 occurs at a point of contact between the load bearing portion 545 and the permanent magnet 518, and is the only substantial stress concentration. (30) FIG. 6 graphically illustrates maximum stress level 452 for the embodiment of the laminate plate 510 and rotor described with reference to FIG. 5 in relation to the maximum stress level 452 for the known laminate plate 410 and rotor described with reference to FIG. 4. The results of FIG. 6 indicate a substantial reduction in the magnitude of the maximum rotor stress by employing the laminate plate 510, as compared to the known laminate plate 410. In one embodiment, the laminate plate 510 and rotor has a reduction of 32% in the maximum rotor stress as compared to the known laminate plate 410 and rotor.

(31) The concepts described herein provide a novel corner geometry that removes material beneath the corner that is away from the magnet. This creates larger surface area that distributes stress and adds dimensional clearance for ease of manufacturing and assembly.

(32) The detailed description and the drawings or figures are supportive and descriptive of the present teachings, but the scope of the present teachings is defined solely by the claims. While some of the best modes and other embodiments for carrying out the present teachings have been described in detail, various alternative designs and embodiments exist for practicing the present teachings defined in the appended claims.

Claims

1. A laminate plate for a rotor of an interior permanent magnet (IPM) electric machine, comprising: a disk-shaped plate defining a plurality of elongated apertures; wherein each of the elongated apertures has a first side and a second side forming opposed first and second sides, and a first end and a second end forming opposed first and second ends; wherein the first end is disposed towards an axis of rotation of the disk-shaped plate, and wherein the second end is disposed towards an outer periphery of the disk-shaped plate; wherein the second end includes a corner portion and a load bearing portion; wherein the load bearing portion is arranged perpendicular to the first side; wherein the corner portion has an edge that defines a void area that is recessed into the first side; wherein the edge includes a convex portion having a first radius R1 arranged in series with a concave portion having a second radius R2, and wherein the first radius R1 is greater than the second radius R2; wherein the convex portion is connected to the first side; wherein the concave portion is connected to the load bearing portion of the second end; and wherein the edge minimizes a stress concentration in the corner portion that is induced by centrifugal force when the rotor is rotating in the IPM electric machine.

2. The laminate plate of claim 1, wherein the edge of the corner portion comprises the convex portion having a first radius R1 and the concave portion having a second radius R2; and wherein the first radius R1 is less than 2 mm, and wherein the second radius R2 is greater than 0.05 mm.

3. The laminate plate of claim 1, wherein the edge of the corner portion comprises the convex portion and the concave portion being a complex spline curve having maximum depth into the disk-shaped plate that is less than 2 mm.

4. The laminate plate of claim 1, wherein the edge minimizes the stress concentration in the corner portion that is induced by centrifugal force of an element that is disposed in the aperture when the rotor is rotating in the IPM electric machine, and wherein the element is arranged to act upon the load bearing portion.

5. The laminate plate of claim 1, wherein the laminate plate is fabricated from electrical steel.
6. The laminate plate of claim 1, wherein the plurality of elongated apertures are circumferentially arranged.
7. A rotor for an interior permanent magnet (IPM) electric machine, the rotor comprising: a rotatable shaft, disposed on a longitudinal axis; a plurality of disk-shaped plates arranged on the rotatable shaft; and a plurality of permanent magnets; wherein each of the plurality of disk-shaped plates defines a plurality of apertures; wherein the plurality of disk-shaped plates are arranged with the plurality of apertures being aligned to form a plurality of axially-disposed cavities; wherein the plurality of permanent magnets are disposed in the plurality of axially-disposed cavities; wherein each of the plurality of apertures of each of the plurality of disk-shaped plates has a first side and a second side forming opposed first and second sides, and a first end and a second end forming opposed first and second ends; wherein the first end is disposed towards the longitudinal axis, and wherein the second end is disposed towards an outer periphery of the disk-shaped plate; wherein the second end includes a corner portion and a load bearing portion; wherein the load bearing portion is arranged perpendicular to the first side; wherein the corner portion has an edge that defines a void area that is recessed into the first side; wherein the edge includes a convex portion having a first radius $R1$ arranged in series with a concave portion having a second radius $R2$, and wherein the first radius $R1$ is greater than the second radius $R2$; wherein the convex portion is connected to the first side; wherein the concave portion is connected to the load bearing portion of the second end; and wherein the edge minimizes a stress concentration in the corner portion that is induced by centrifugal force when the rotor is rotating.
8. The rotor of claim 7, wherein the edge of the corner portion comprises the convex portion having a first radius $R1$ and the concave portion having a second radius $R2$; wherein the first radius $R1$ is less than 2 mm, and wherein the second radius $R2$ is greater than 0.05 mm.
9. The rotor of claim 7, wherein the edge of the corner portion comprises the convex portion and the concave portion being a complex spline curve having maximum depth into the disk-shaped plate that is less than 2 mm.
10. The rotor of claim 7, wherein the edge minimizes the stress concentration in the corner portion that is induced by centrifugal force when the rotor is rotating in the IPM electric machine, and wherein the permanent magnet is arranged to act upon the load bearing portion.
11. The rotor of claim 7, wherein the disk-shaped plate is fabricated from electrical steel.
12. The rotor of claim 7, wherein the plurality of apertures are circumferentially arranged.
13. An interior permanent magnet (IPM) electric machine, comprising: a rotor rotatably arranged within a stator; the rotor comprising: a rotatable shaft, disposed on a longitudinal axis; a plurality of laminate plates arranged on the rotatable shaft; and a plurality of permanent magnets; wherein each of the plurality of laminate plates defines a plurality of circumferentially arranged apertures; wherein the plurality of laminate plates are arranged with the plurality of circumferentially arranged apertures being aligned to form a plurality of axially-disposed cavities; wherein the plurality of permanent magnets are disposed in the plurality of axially-disposed cavities; wherein each of the apertures of the plurality of laminate plates has a first side and a second side forming opposed first and second sides, and a first end and a second end forming opposed first and second ends; wherein the first end is disposed towards an axis of rotation of the plurality of laminate plates; wherein the second end is disposed towards an outer periphery of the plurality of laminate plates; wherein the second end includes a corner portion and a load bearing portion; wherein the corner portion has an edge that defines a void area that is recessed into the first side; wherein the edge includes a convex portion having a first radius $R1$ arranged in series with a concave portion having a second radius $R2$, and wherein the first radius $R1$ is greater than the second radius $R2$; wherein the convex portion is connected to the first side; wherein the concave portion is connected to the load bearing portion of the second end; and wherein the edge minimizes a stress concentration in the corner portion that is induced by centrifugal force when the rotor is rotating in

the stator.

14. The IPM electric machine of claim 13, wherein the edge of the corner portion comprises the convex portion having a first radius $R1$ and the concave portion having a second radius $R2$; wherein the first radius $R1$ is less than 2 mm, and wherein the second radius $R2$ is greater than 0.05 mm.

15. The IPM electric machine of claim 13, wherein the edge of the corner portion comprises the convex portion and the concave portion being a complex spline curve having maximum depth into the plate that is less than 2 mm.

16. The IPM electric machine of claim 13, wherein the edge minimizes the stress concentration in the corner portion that is induced by centrifugal force when the rotor is rotating in the stator, and wherein the one of the plurality of permanent magnets is arranged to act upon the load bearing portion.

17. The IPM electric machine of claim 13, wherein the load bearing portion is arranged perpendicular to the first side.
